

BRL Architecture

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Abstract

In this paper, we present a knowledge-based POMDP framework that enables robots to selectively request human feedback under uncertainty. While modern robotic systems achieve strong autonomy, they still fail due to perception errors, execution failures, and environmental changes, often requiring continuous human supervision. We model decision making as a POMDP over a symbolic knowledge base, where uncertainty is quantified using the entropy of a belief over possible world states to determine when to query. The system also determines what to query by selecting informative queries that reduce ambiguity between competing hypotheses. We evaluate the proposed method in both a simulated waste sorting task and a real-world tomato harvesting task. Results show that our approach maintains task success rates comparable to full supervision while significantly reducing the number of queries, and user studies confirm reduced cognitive workload during interaction.

Keywords: robotic system, planning under uncertainty, human-robot interaction

1 Introduction

Recent advances in robotic systems have significantly improved autonomy in perception, manipulation, and planning across domains such as agriculture and household environments. Despite these advances, robots still encounter perception errors [1], action execution failures [2], and unexpected environmental changes [3]. As a result, many robotic systems continue to rely on human operators to monitor execution and

intervene when failures occur [4, 5]. However, requiring humans to continuously monitor robotic systems and determine whether robot decisions are correct is impractical in long-horizon tasks because it increases cognitive workload and fatigue [6]. Instead of relying on human operators to detect failures after they occur, robotic systems should be able to recognize uncertain situations during execution and request assistance only when additional information is required [7, 8]. This raises two fundamental questions: (i) when should a robot request human assistance, and (ii) what information should it request?

Recent studies have explored uncertainty-aware robotic systems that selectively request human feedback during task execution [9–11]. These approaches demonstrate that uncertainty-aware querying can reduce unnecessary human intervention and improve decision making under uncertainty. However, uncertainty estimation, query generation, and planning are often treated as separate components. As a result, the relationship between when to request human assistance, what information to request, and how the acquired feedback should influence subsequent decision making remains insufficiently explored in robotic systems operating under uncertainty.

To address this challenge, we propose a knowledge-based robotic framework that integrates uncertainty estimation, query selection, and task planning into a unified decision-making process. We formulate decision making as a partially observable Markov decision process (POMDP) [2] and represent task knowledge using a STRIPS-style symbolic model [12, 13]. The system maintains a belief over alternative world hypotheses and continuously updates this belief using observations collected during execution. The resulting belief distribution is used to determine whether additional information is required. When uncertainty exceeds a threshold, the robot requests feedback that is expected to maximally reduce ambiguity among competing hypotheses. Otherwise, the robot continues autonomous planning and execution.

To support autonomous operation under uncertainty, we integrate the proposed framework into a robotic system architecture. As illustrated in Fig. 1, a centralized knowledge base connects the sensing, planning, feedback management, and user interface modules. Sensor observations and user feedback are stored in the knowledge base and shared across modules, providing a consistent representation of the environment throughout execution. Planning is performed over symbolic world states, while belief updates maintain uncertainty over alternative hypotheses. Consequently, the robot can continue autonomous execution and request human feedback only when necessary.

We evaluate the proposed approach in two robotic domains, tomato harvesting and waste sorting. Experimental results demonstrate that the framework improves task performance while reducing the amount of human intervention required during execution. We further analyze how different query policies influence situation awareness and perceived workload through user studies.

The main contributions of this work are as follows:

- We propose an uncertainty-aware feedback triggering method for determining when human assistance is required during task execution.
- We propose an information-driven query selection method for identifying what information should be requested to reduce uncertainty over competing world hypotheses.

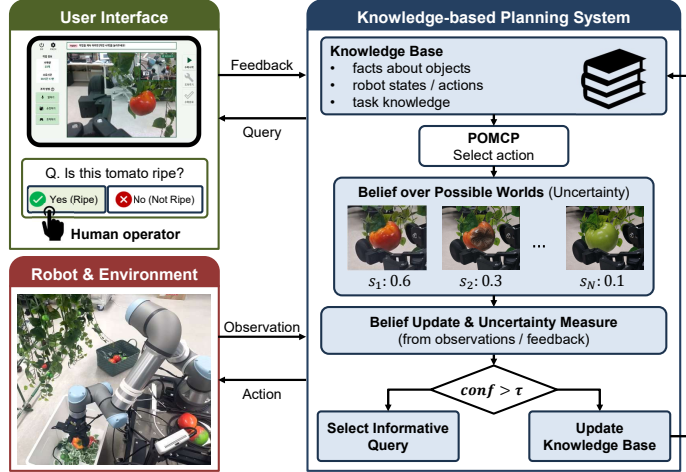


Fig. 1: Overview of the proposed knowledge-based robotic system with uncertainty-aware planning. The system maintains a belief over possible worlds and updates it based on observations. If the uncertainty exceeds a threshold, the system selectively queries the user to resolve ambiguity. Otherwise, it continues autonomous planning and execution.

- We incorporate uncertainty-aware querying into the planning and execution process, enabling autonomous robot operation with selective human feedback.
- We validate the proposed approach in tomato harvesting and waste sorting tasks through robot experiments and user studies.

2 Related Work

In this section, we review related work along three directions. We first discuss knowledge-based robotic systems for structured environment representation. We then review planning under uncertainty, including POMDP-based and belief space methods. Finally, we examine human corrective systems that incorporate feedback for decision making.

2.1 Knowledge-based Robotic System

In robotics, a knowledge base refers to a structured representation of information required for decision making. These systems use logic-based representations of objects, attributes, and relations to describe the world in a way that is understandable not only by robots but also by users [64, 65]. In particular, predicate-based symbolic representations such as [13, 24] allow the system to describe both object properties and relationships among objects. In addition, graph-based models such as [66, 67] are often used to represent these relationships more clearly by organizing objects and their connections in a simple structured form.

Table 1: Qualitative comparison of related works.

Related Works	Symbolic KB	Symbolic Planning	Planning under Uncertainty	Human Feedback	Proactive Query
Logic-based knowledge inference [14–23]	○	×	△	×	×
Classical planning [13, 24–28]	○	○	×	×	×
LLM-based knowledge inference [29–35]	○	△	×	×	×
LLM-based planning [36–45]	○	△	△	△	×
POMDP-based planning [46–54]	×	△	○	×	×
Symbolic planning under uncertainty [55–59]	○	○	○	×	×
Human-on-the-loop systems [60–63]	×	×	△	○	×
Uncertainty-aware query systems [9–11]	△	×	○	○	○
Ours	○	○	○	○	○

To systematically define robot states and actions, planning languages such as PDDL [24] or FDR [25] are widely adopted. In these representations, a state is defined as a set of predicates, and each action is described by its preconditions and effects, which specify how the state changes after execution. This formulation provides a clear model of state transitions and allows robots to plan over a well-defined symbolic state space. As a result, classical planning techniques [26–28] can be applied to compute action sequences that achieve a given goal.

Knowledge-based robotic systems store observed facts and infer new knowledge from existing information within this symbolic representation. Logical and relational inference allows robots to identify facts that are not directly observed, combine multiple relations, and obtain plausible hidden states from contextual knowledge [14, 15]. For example, a robot can infer the likely location of an unseen object from qualitative spatial relations, container-location relations, or place-object co-occurrence knowledge [16, 17]. Commonsense reasoning connects symbolic knowledge with perceptual evidence and supports decision making in partially observable environments [18–20]. Ontology structures and type hierarchies further support consistent integration of new observations and generalization across related objects, actions, and task contexts [21–23].

While symbolic knowledge-based systems provide structured representations and support logical inference, recent approaches have explored the use of large language models (LLMs) to further expand how knowledge is utilized in robotic planning. Early LLM-based approaches employ procedural and commonsense knowledge encoded in LLMs to propose task decompositions or action sequences. The proposed plans are then grounded with robot affordances, programmatic interfaces, or execution feedback [29–33]. Recent LLM-based approaches also incorporate external knowledge sources to compensate for incomplete observations. Some methods use scene graphs

or open-vocabulary spatial representations to provide structured information about objects, places, and relations [34, 35]. Other methods retrieve demonstrations, procedural knowledge, or task-relevant constraints to guide long-horizon planning [36–40]. Related methods combine LLM-generated plans with symbolic constraints, spatial reasoning, or verification mechanisms to improve executability and consistency [41–45].

However, many knowledge-based robotic systems still rely on deterministic symbolic representations and assume a single consistent world state, which limits their ability to represent multiple hypotheses under uncertainty [68, 69]. As a result, such systems do not explicitly model uncertainty in action effects, sensing, or world states, and inferred facts are typically committed as definite knowledge. Recent LLM-based planning methods also suffer from grounding errors, incomplete retrieval, and hallucinated inferences, and do not maintain structured uncertainty over alternative states. While knowledge-based systems provide rich semantic structure for planning, they do not directly address uncertainty in observations, inferred facts, and knowledge validity.

To address this limitation, we maintain multiple symbolic hypotheses over possible worlds instead of committing inferred facts as deterministic knowledge. Our framework combines symbolic knowledge-based reasoning with planning under uncertainty, allowing the robot to reason over uncertain inferred states while selectively requesting human feedback if the uncertainty becomes high.

2.2 Planning under Uncertainty

2.2.1 POMDP

Unlike deterministic symbolic knowledge-based systems that assume a single world state, POMDPs explicitly reason over uncertainty through belief representations over multiple possible states. In robotic environments, such uncertainty naturally arises from sensor noise, occlusions, incomplete observations, and unexpected state changes. Sequential decision-making under these conditions has been extensively studied within the framework of POMDPs [46–49].

A key distinction between a POMDP and an MDP is the introduction of a belief state, which represents a probabilistic distribution over possible states when the true state cannot be directly observed. Accordingly, a wide range of solution methods have been proposed, including point-based value iteration [70], online tree search [71], and Monte Carlo tree search (MCTS) [72].

Despite these advances, effectively representing and updating the belief state remains a fundamental challenge. Since a belief is inherently a continuous distribution over the state space, exact inference is computationally difficult. Moreover, as the state space grows, the belief space expands combinatorially, making inference increasingly intractable. As a result, various approaches have been proposed to approximate the belief state.

2.2.2 Belief Approximation

Studies on belief approximation can be broadly categorized into particle-based and factored-based. Particle-based methods approximate the belief as a set of sampled states and have been widely adopted due to their natural compatibility with sampling-based online planning [73, 74]. For example, [72] maintains a set of particles as a history-conditioned belief and estimates action values by sampling future trajectories from a generative model without performing exact Bayesian belief updates. Similar sampling-based ideas have also been extended to large and continuous POMDPs, where DESPOT [46] evaluates a limited set of sampled scenarios and POMCPOW [75] controls search tree growth in continuous action and observation spaces.

In robotic domains, particle-based belief representations are widely used in localization, navigation, SLAM, object search, and manipulation, where particles encode hypotheses over robot, map, and object states under uncertainty. In navigation, [50] combines POMCP with model predictive control to maximize information gain over occupancy-grid maps, while [51] uses hierarchical local online POMDP planning for exploration in large unknown environments. Similar representations have also been used for continuous control under uncertainty, including autonomous driving at occluded intersections [76] and UAV collision avoidance or target tracking under partial observability [52, 77]. In object search and manipulation, particles often represent complete scene hypotheses. For example, [53] samples visible and occluded object configurations for cluttered object search, while [54, 78] uses particle-based scene hypotheses for multi-object manipulation. Collectively, these studies show that particle beliefs provide a flexible approximation for preserving multimodal dependencies among robot, environment, and object states.

In contrast, factored belief representations model the belief by decomposing the state into a set of variables and exploiting their structure [79–81]. Such approaches have been widely used in structured probabilistic models, including dynamic Bayesian networks and Hidden Markov Models [82], where the state is represented in a factored form. To improve computational efficiency, various approximate inference methods have been proposed to reduce the complexity of belief updates by relaxing dependencies among variables. For example, [83] approximates belief updates by maintaining a factorized representation while partially accounting for dependencies between variables. While factored representations exploit internal structure of state space, their independence assumptions break in symbolic planning where facts are tightly linked by action constraints. Therefore, we use a particle-based approximation to preserve these dependencies.

2.2.3 Symbolic Planning under Uncertainty

Symbolic planning formalisms such as PDDL [24], STRIPS [13], and FDR [25] have been widely used due to their interpretability [84]. These representations are typically based on first-order logic or similar discrete formalisms [85] which describes planning problems through explicit constraints and state transition rules. However, since traditional predicate logic only allows binary true or false values, it is difficult to directly model observation noise or state uncertainty. To address this limitation,

various approaches have attempted to integrate probabilistic reasoning into logical representations. For example, [86] assign weights to logical rules to enable probabilistic inference over uncertain relationships, and [87] combine logical structure with probabilistic reasoning.

However, these approaches primarily focus on building inference models, and their high cost in model construction is unsuitable for online decision-making with complex constraints.

As a result, several studies have explored planning under uncertainty while preserving symbolic structure [55, 56]. For example, [57] encodes commonsense knowledge in ASP [85] to construct a prior over possible worlds and integrates it with probabilistic planning [88]. However, since the belief is defined over a set of possible worlds, the computational cost increases as the number of states grows. To address this, some works organize possible worlds in a structured manner. For instance, [58] construct a state space tree and a hierarchy of POMDPs based on domain knowledge, reducing complexity by leveraging abstraction. More recently, large language models have been used for belief approximation, as in [59], which builds a tree of hypotheses and uses LLMs to generate particles in open-ended settings. While these approaches enable more flexible or scalable belief representations, they either depend on predefined hierarchical structures or rely on the quality and consistency of generated hypotheses.

Prior work has explored various belief representations, but global belief approaches remain sample-inefficient under strong symbolic constraints. We instead approximate belief locally by sampling states reachable under each action [88]. These local beliefs are organized in a tree structure, and planning is performed using the off-the-shelf planner POMCP within an interpretable symbolic space. However, uncertainty in real-world robotic systems cannot always be resolved through planning and observation alone. Robots may therefore require additional information to disambiguate conflicting hypotheses, which motivates human corrective systems that selectively request feedback under high uncertainty.

2.3 Human Corrective System

Despite recent advances in robotic systems, robots still face operational uncertainty in dynamic and unpredictable real-world environments. Prior work has addressed this limitation by involving humans in different forms of robot autonomy [62, 63]. In particular, in human-on-the-loop (HOTL) systems, robots execute tasks autonomously, while human operators monitor the system and revise or override robot decisions if failures or errors occur [60, 61]. This supervisory structure can improve safety and robustness because human operators can compensate for errors that autonomous systems cannot resolve.

However, HOTL systems require operators to sustain attention and detect failures during robot operation. To provide effective corrective feedback, operators must first notice that the robot is in an uncertain or erroneous state. Situation awareness research suggests that operators must then understand the current situation, project possible consequences, and determine an appropriate intervention [89]. This cognitive process can be difficult in repetitive tasks. Operators may experience vigilance decrement because continuous exposure to monotonous monitoring tasks can slow responses

and reduce detection accuracy [90, 91]. Operators may also experience inattentive blindness, in which visible failures are missed if attention is not directed to the relevant event [92]. As a result, continuous supervision can increase workload and fatigue [93]. Continuous supervision can also reduce the likelihood that operators detect critical robot failures [93, 94].

Prior work has shown that robots can improve task performance, team efficiency, and subjective satisfaction if they proactively provide assistance during human-robot collaboration [95]. This finding suggests that proactive robot behavior can support collaboration by reducing the need for humans to manage every decision or detect every need for intervention. In parallel, uncertainty-aware robot systems have advanced toward determining when human input is needed. For example, conformal prediction-based methods construct prediction sets with statistical guarantees and use the size of these sets to decide whether to query the user [9–11]. These methods show that robots can use uncertainty estimates to reduce unnecessary human intervention.

However, studies that develop and evaluate system-initiated query systems have mainly focused on technical performance, such as task performance, uncertainty calibration, and query efficiency. Human factors, such as workload and fatigue, can also affect task performance in automated systems that involve human operators [96–98]. Therefore, system-initiated corrective interaction should be evaluated not only in terms of technical performance but also in terms of operator burden and fatigue.

Building on this gap, we examine how system-initiated corrective interaction affect operator workload, fatigue, and task performance in automated robot systems. Our system does not require users to continuously supervise the robot to detect failures. Instead, users provide corrective feedback only if the system requests input. This interaction structure shifts part of the problem detection burden from the user to the robot.

3 Preliminaries

3.1 Symbolic State and Action Representation

The objective of planning is to find a transition sequence that transforms the initial state into a goal state. The planning problem is defined as a tuple $\mathcal{P} = \langle \mathcal{F}, \mathcal{A}, s_0, \mathcal{G} \rangle$, where $\mathcal{F}$ is a finite set of **atoms**, $\mathcal{A}$ is a set of actions, $s_0 \subseteq \mathcal{F}$ is the initial state, and $\mathcal{G} \subseteq \mathcal{F}$ is the goal condition. A state s is represented as a subset of $\mathcal{F}$. Each action $a \in \mathcal{A}$ is defined as a tuple $a = \langle \text{pre}(a), \text{add}(a), \text{del}(a) \rangle$, where $\text{pre}(a), \text{add}(a), \text{del}(a) \subseteq \mathcal{F}$ denote the preconditions, add lists, and delete lists of the action, respectively. **An action a is said to be applicable in state s if and only if $\text{pre}(a) \subseteq s$, denoted as $a[s]$.** The state transition is defined as $s \xrightarrow{a} s'$, where $s' = (s \setminus \text{del}(a)) \cup \text{add}(a)$.

A plan is a sequence of actions $\pi = (a_1, \dots, a_T)$ such that the goal condition is satisfied after sequentially applying the actions starting from the initial state, i.e.,

$$s_0 \xrightarrow{a_1} s_1 \xrightarrow{a_2} \dots \xrightarrow{a_T} s_T \quad \text{and} \quad \mathcal{G} \subseteq s_T.$$

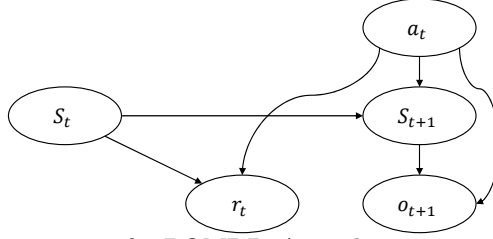


Fig. 2: Graphical illustration of a POMDP. At each time step t , the agent selects an a_t based on its current belief over the latent state s_t . The environment transitions to the next state s_{t+1} and generates a reward r_t . Since the true state is not directly observable, the agent receives an observation o_{t+1} .

3.2 POMDP

A POMDP is defined as a tuple $\langle \mathcal{S}, \mathcal{A}, T, R, \Omega, O, \gamma \rangle$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, and Ω is the observation space. The transition model $T(s' | s, a)$ specifies the probability of reaching the next state s' after taking action a in current state s . The observation model $O(o | s', a)$ defines the probability of receiving observation o after arriving at state s' by executing action a . The reward function $R(s, a)$ gives a scalar value to each transition, and $\gamma \in (0, 1]$ is the discount factor.

Since the true state is not directly observable, the agent maintains a belief b , which is a probability distribution over $\mathcal{S}$ such that $b(s) = \Pr(s)$. After executing an action a and receiving an observation o , the belief is updated using a Bayes filter.

$$b'(s') = \eta O(o | s', a) \sum_{s \in \mathcal{S}} P(s' | s, a) b(s), \quad (1)$$

where η is a normalization constant.

A policy π maps beliefs to actions. The objective of a POMDP is to find a policy that maximizes the expected cumulative discounted reward, defined as $\mathbb{E} [\sum_{t=0}^{\infty} \gamma^t r_t]$.

3.3 POMCP

Partially Observable Monte Carlo Planning (POMCP) performs Monte Carlo tree search over action-observation histories $h = \langle a_1 o_1 \dots a_t o_t \rangle$, where each node in the search tree corresponds to a history. Action selection follows the UCB rule:

$$a^* = \arg \max_a \left(V(ha) + c \sqrt{\frac{\log N(h)}{N(ha)}} \right), \quad (2)$$

where $V(ha)$ is the estimated value of a after h , $N(h)$ and $N(ha)$ denote visit counts, and c is an exploration constant.

Each simulation samples the next state, observation, and reward from a generative model $(s', o, r) \sim \mathcal{G}(s, a)$. The return is computed recursively as $R = r + \gamma R'$,

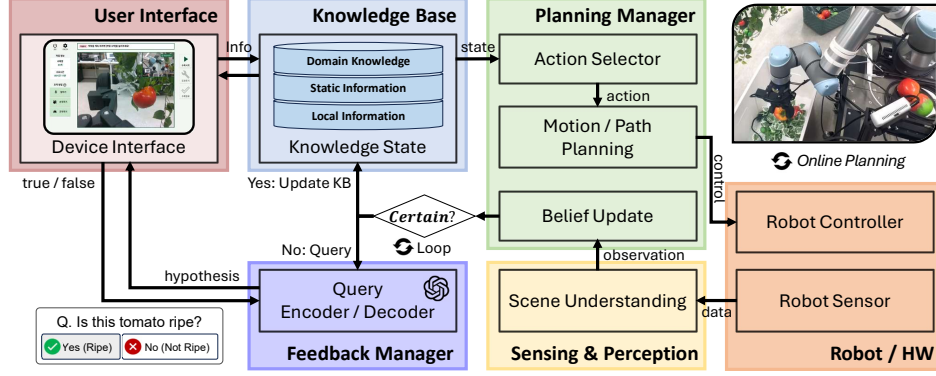


Fig. 3: System architecture and data flow of the proposed framework. The system consists of the *Knowledge Base* (KB), *Planning Manager* (PM), *Sensing and Perception* module (SP), *Feedback Manager* (FM), and *User Interface* (UI). The PM performs online planning based on the current knowledge state stored in the KB and updates the belief using observations from SP. The system evaluates the confidence of the updated belief. If the confidence exceeds a threshold, the most probable hypothesis is committed to the KB. Otherwise, the FM generates a query and requests corrective feedback from the user through the UI. The feedback is then used to refine the belief and continue planning.

where R' is the return from subsequent steps. The value estimate is updated using an incremental average:

$$V(ha) \leftarrow V(ha) + \frac{R - V(ha)}{N(ha)}. \quad (3)$$

The belief associated with a history h is represented implicitly as a set of particles, $B(h) = \{s \mid s \text{ is visited at } h\}$, which approximates the posterior over states without explicit belief updates.

4 System Architecture

In this section, we present an overview of the system architecture and the data flows between modules. Then we give a detailed explanation of their functionalities and applications.

4.1 Overview

Fig. 3 illustrates the brief system architecture and the data flow between the main modules. We integrate the *Knowledge Base* (KB), the *Planning Manager* (PM), the *Sensing and Perception* module (SP), the *Feedback Manager* (FM), and the *User Interface* (UI). The KB maintains symbolic descriptions and facts used for planning, SP generates sensory information from robot sensors, PM generates and executes plans, FM coordinates corrective feedback, and UI bridges the system and the user.

The KB stores essential information required for the robot to perform tasks and explicitly represents the current state of the system. All modules share information

and interact with each other through the KB. The KB is composed of three types of information: domain knowledge, static information, and local information. Domain knowledge refers to prior knowledge in a specific domain. It includes descriptions of robot skills, domain-specific rules, and general explanations of the environment. Static information is used during planning and execution but remains unchanged throughout task execution. Unlike domain knowledge, static information is encoded in a structured form and includes information such as robot parameters and local maps. In contrast, local information is used during planning and execution but cannot be known in advance. It is obtained through observations from the SP module and stored in the KB in the form of facts.

The PM is responsible for online action selection and execution under partial observability. The POMCP-based planner within the PM module selects the best action from the current belief and triggers the SP module to acquire additional observations. For instance, if the robot is positioned in front of a tomato stem, it naturally executes a **detect** action to infer the locations and ripeness states of tomatoes. To quantify uncertainty, the PM approximates the belief in the form of particle and computes the entropy over the resulting belief distribution. If such uncertainty is detected, the PM queries the FM to obtain additional information that can resolve the ambiguity of the observation. Upon receiving a response from the FM, the PM updates the belief accordingly and promotes the most probable belief state to a certain state in the knowledge base.

The SP module provides the current visual sensing information to the UI in real time. In addition, it updates the KB with observations triggered by the PM. In this process, the SP module makes use of data from various sensors, such as cameras, LiDAR, and tactile sensors, to extract information used for task execution, including object locations, states, and properties. For example, from RGBD images, the SP module infers visual attributes such as the position, ripeness, and state of a tomato, while tactile sensors are used to perceive contact-based properties such as object stiffness or softness. The high-level perceptual information extracted by the SP module is triggered by the BT and updated in the KB in the form of symbolic facts.

The FM module is implemented using an LLM-based interface that translates symbolic hypotheses from the PM into user-friendly natural language queries. For example, a hypothesis such as `ripe(tomato1)` is converted into a question like "Is this tomato ripe?". Along with the query, the module provides corresponding visual cues in the user interface, such as highlighting the referenced object with a bounding box. The FM module also interprets user interactions with the interface (e.g., button clicks) and maps them to boolean feedback, which is then incorporated back into the system as validated truth assignments.

Finally, the UI displays the current visual sensing information and the system state, and allows the user to provide direct feedback or make selections. The user interacts with the FM module through the UI to monitor and supervise task execution.

4.2 Modules

4.2.1 Knowledge Base

The KB maintains a structured representation of the environment, including objects, properties, task-relevant relations, numeric quantities, and their associated confidence values. Formally, we define $\mathcal{KB} = \mathcal{I}^D \sqcup \mathcal{I}^S \sqcup \mathcal{I}^L$,¹ where $\mathcal{I}^D$, $\mathcal{I}^S$, and $\mathcal{I}^L$ denote the domain, static, and local information sets, respectively, as described in Sec. 4.1.

For, $\mathcal{I}^S$ and $\mathcal{I}^L$, each information set contains both symbolic facts and numeric fluents. A symbolic fact is defined as $f = \text{pred}(\mathbf{x})$, where $\text{pred}(\cdot)$ denotes a predicate symbol and $\mathbf{x}$ denotes its arguments. A numeric fluent is defined as $\nu = \langle g(\mathbf{x}), v \rangle$, where $g(\mathbf{x})$ denotes a numeric function over objects and $v \in \mathbb{R}$ denotes its value. For example, symbolic facts can represent object categories and relations, such as `tomato(tomato1)` and `loc(stem1)`, while numeric fluents can represent measurable quantities, such as `pose_x(tomato1) = 0.12` or `distance(robot, stem1) = 2.5 (m)`.

Specifically, the set $\mathcal{I}^S$ contains information that remains invariant throughout planning. By definition, for all $a \in \mathcal{A}$, $\text{Effect}(a) \cap \mathcal{I}^S = \emptyset$, where $\text{Effect}(a) = \text{add}(a) \cup \text{del}(a)$ denotes symbolic effects and can be extended to include numeric updates. These elements capture structural properties of the environment that are not altered by any action. For example, predicates such as `tomato(tomato1)` and `loc(stem1)`, as well as fixed numeric quantities like `loc_x(stem1) = 1.2`, remain unchanged regardless of the executed actions.

In contrast, $\mathcal{I}^L$ denotes local information obtained through the SP module and is not known a priori. This includes perceptual predicates, such as object existence or category estimates, as well as numeric fluents estimated from sensor data. For example, predicates such as `ripe(tomato1)` or `detected(tomato2)` may be inferred from observations, and numeric values such as `distance(robot, stem1) = 2.3` or `pose_x(tomato1) = -0.5` are updated as new sensor measurements become available.

4.2.2 Planning Manager

The PM performs decision making based on the current belief state while continuously updating the knowledge base during execution. Uncertainty is quantified using the entropy of the belief distribution. If the uncertainty exceeds a threshold, the planner requests additional verification from the FM. Otherwise, it commits the most probable hypothesis to the knowledge base and proceeds with action execution. In the following, we describe the belief approximation, the uncertainty estimation, the POMCP based search, and the overall planning procedure.

Belief representation

In partially observable environments, the true state cannot be directly observed, and thus the PM maintains a belief, which is a probability distribution over states. The belief at time t is defined as $b_t(s) = P(s_t = s \mid h_t)$, where $h_t = \{a_1, o_1, \dots, a_t, o_t\}$ denotes the history of actions and observations. However, in symbolic domains, the state space is large and subject to structural constraints, making exact belief tracking computationally intractable. Thus, we approximate the belief using a set of particles.

¹The notation $\sqcup$ denotes a disjoint union, indicating that $\mathcal{I}^D$, $\mathcal{I}^S$, and $\mathcal{I}^L$ form a partition of $\mathcal{KB}$.

Formally, at time step t , the belief is represented as a finite set of particles and their associated weights:

$$\mathcal{B}_t = \{(s_i, w_i)\}_{i=1}^N, \quad \sum_{i=1}^N w_i = 1, \quad (4)$$

where each s_i denotes a possible world composed of a set of facts. The approximated belief distribution is then

$$b_t(s) \approx \sum_{i=1}^N w_i \delta(s_i = s). \quad (5)$$

In the symbolic domain, actions exhibit partially predictable effects due to their interpretable structure. For example, `pick(tomato1)` may result in grasp failure or selecting a different tomato, but it does not plausibly change the location of the robot. Based on this observation, we define a restricted set of reachable successor states, referred to as the *frontier*. For each s and a , the frontier is defined as

$$Frontier(s, a) = \{s_1^*, s_2^*, \dots, s_j^*\}, \quad (6)$$

where each s_k^* denotes a reachable successor state.

In contrast to standard particle filtering, we assume a *knowledge* state s^K that represents the set of certain facts. Given a and s^K , we construct the belief by considering only the reachable states induced by s^K . That is, particles are defined exclusively over the frontier $Frontier(s^K, a)$, and the belief is represented as

$$\mathcal{B}_t = \{(s_k^*, w_k)\}_{k=1}^j, \quad \sum_{k=1}^j w_k = 1, \quad (7)$$

where each s_k^* is a reachable successor state such that $s_k^* \in Frontier(s^K, a)$. After executing a and receiving an observation o , the belief over the frontier is updated by adjusting the weight of each particle according to how well it explains the observation. Specifically,

$$\forall s_k^* \in Frontier(s^K, a), \quad w'_k = w_k \cdot O(o \mid s_k^*). \quad (8)$$

followed by normalization to obtain the *uncertainty* of the current state.

Uncertainty of the state

Given the updated belief over the frontier, uncertainty is quantified based on the distribution of particle weights. Intuitively, uncertainty increases when multiple candidate states have comparable likelihoods, and decreases when the belief is concentrated on a single dominant state. This uncertainty arises from *hypotheses*, which denote literals whose truth values are not yet determined.

The confidence of the current belief is defined as

$$\text{Conf}(\mathcal{B}) = 1 - \frac{\mathcal{H}(\mathcal{B})}{\log_2 j}, \quad (9)$$

where j denotes the number of particles. In the degenerate case where $j = 1$, the belief is fully concentrated on a single state and we define $\text{Conf}(\mathcal{B}) = 1$. Here, $\mathcal{H}(\mathcal{B})$ denotes the entropy of the normalized particle weights. Let $\tilde{w}'_k$ denote the normalized weights such that $\tilde{w}'_k = w'_k / \sum_{l=1}^j w'_l$. The entropy is computed as

$$\mathcal{H}(\mathcal{B}) = - \sum_{k=1}^j \tilde{w}'_k \log_2 \tilde{w}'_k. \quad (10)$$

This normalization ensures that the confidence lies in $[0, 1]$, where a uniform belief yields low confidence and a concentrated belief yields high confidence.

If the confidence exceeds a predefined threshold τ , the most probable state is selected as the updated knowledge, $s^* = \arg \max_{s_k} w'_k$, and the belief collapses to this state by discarding all remaining particles. Otherwise, the system identifies uncertainty-inducing hypotheses that differentiate candidate states.

The candidate hypotheses are constructed from literals that differ between the current knowledge state and the frontier states. Let $\mathcal{C}$ denote this set of candidate hypotheses. For each $c \in \mathcal{C}$, the belief is partitioned into two subsets depending on whether c holds:

$$\mathcal{B}_c^+ = \{(s_k^*, w'_k) \mid c \in s_k^*\}, \quad \mathcal{B}_c^- = \{(s_k^*, w'_k) \mid c \notin s_k^*\}.$$

The system selects the hypothesis that is expected to reduce the uncertainty the most by minimizing the expected entropy after querying:

$$c^* = \arg \min_{c \in \mathcal{C}} [P(c) \mathcal{H}(\mathcal{B}_c^+) + P(\neg c) \mathcal{H}(\mathcal{B}_c^-)], \quad (11)$$

where

$$P(c) = \sum_{s_k^*: c \in s_k^*} w'_k \quad \text{and} \quad P(\neg c) = \sum_{s_k^*: c \notin s_k^*} w'_k.$$

The selected hypothesis c^* is passed to the FM as a query. After receiving the answer, the belief is filtered by removing particles that contradict the feedback. If the answer confirms c^* , only particles containing c^* are retained. Otherwise, particles containing c^* are discarded:

$$\mathcal{B} \leftarrow \begin{cases} \{(s_k^*, w'_k) \mid c^* \in s_k^*\}, & \text{if } ans = \text{true}, \\ \{(s_k^*, w'_k) \mid c^* \notin s_k^*\}, & \text{if } ans = \text{false}. \end{cases} \quad (12)$$

The remaining weights are renormalized, and the confidence is recomputed. This process is repeated until the confidence exceeds τ or no informative hypothesis remains.

POMCP-based search

We employ POMCP [72] as the underlying planner for action selection. We use its sampling-based framework and focus on integrating symbolic representations with uncertainty-aware reasoning. Instead, as summarized in Alg. 2 to Alg. 4, we apply

minimal adaptations to ensure consistency with the symbolic domain. The detailed pseudocode for the search, simulation, and action-selection routines is provided in Appendix A.

In Alg. 2, the search phase follows standard POMCP by repeatedly sampling a state and performing simulations. Each sampled state is passed to **SIMULATE** (line 9), and the accumulated values are used to select the final action. If the belief is empty, the search uses s^K , and at the root, the action is chosen based on value without UCB (line 11).

In the simulation phase, as described in Alg. 3, the recursive structure and return computation remain the same as in standard POMCP, but both action selection and node expansion are restricted to $\mathcal{A}(s)$ (line 8). That is, instead of considering the full action space, only actions applicable in state s are explored. If a new history h is encountered, the node is expanded only with actions in $\mathcal{A}(s)$ (line 9), and the value is initialized via a **ROLLOUT**. Here, T denotes the search tree, while $N(ha)$ and $V(ha)$ represent the visit count and value of action a , respectively. The next state, observation, and reward are then sampled from the generative model $(s', o, r) \sim \mathcal{G}(s, a)$ (line 11). As a result, the simulation remains stochastic but is confined to reachable symbolic transitions.

In Alg. 4, the same principle is applied to both action selection and rollout. The candidate set $\mathcal{C} = \{a \mid a[s] \in \mathcal{A}, ha \in T\}$ includes actions that are both applicable in state s and present in the tree. If `use_ucb` is true, an action is selected according to the UCB criterion defined in Eq. 2. Otherwise, the action with the highest value is chosen. During rollout, actions are sampled uniformly from all applicable actions. The generative model $\mathcal{G}(s, a)$ then produces the next state, observation, and reward, while the action space is restricted by symbolic constraints.

In symbolic domains, action preconditions and effects are explicitly defined, so exploring infeasible actions is unnecessary. Restricting the action space to $\mathcal{A}(s)$ therefore improves both search efficiency and model consistency. Overall, the structure follows POMCP while naturally incorporating the constraints imposed by the symbolic representation.

Action execution

Each $a \in \mathcal{A}$ selected by the POMCP is associated with an executable robot skill. The symbolic action specifies the task-level intention and parameters, while the corresponding skill defines the motion or perception routine required for execution. For example, a symbolic action such as `navigate(robot,  $l_1, l_2$ )` is mapped to a navigation routine, and `pick(robot, tomato1)` is mapped to a predefined manipulation trajectory.

In this work, $\forall a \in \mathcal{A}$ follows an encoded execution policy that maps a symbolic operator to a robot skill through an off-the-shelf motion planner, such as MoveIt [99]. The planner takes symbolic facts and numeric fluents stored in the KB as inputs. Numeric fluents, such as object poses, distances, and grasping parameters, define the parameters of motion generation and maintain consistency between high-level planning and low-level execution.

Specifically, after each action execution, the system receives an observation o from the environment and uses it for belief update. The system uses these observations to

Algorithm 1: Planning Pipeline

```
1 Input: initial knowledge  $s_0^K$ , confidence threshold  $\tau$ 
   Output: plan  $\pi$ 
2 procedure PLANNING( $s_0^K$ )
3    $knowledge(\mathcal{B}) \leftarrow s_0^K$ 
    $t \leftarrow 0$ 
4   repeat
5      $a_t \leftarrow \text{SEARCH}(\mathcal{B})$ 
6     if  $a_t = \emptyset$  then
7       // There is no  $a \llbracket s \rrbracket$ . (Dead-end)
       break
8     execute  $a_t$  and receive observation  $o_{t+1}$ 
      $\mathcal{B} \leftarrow \text{UPDATE}(\mathcal{B}, a_t, o_{t+1})$  (Eq. 8)
      $c \leftarrow \text{CONF}(\mathcal{B})$  (Eq. 9)
     while  $c < \tau$  do
9        $hypo^* \leftarrow \text{SELECT-HYPO}(\mathcal{B})$  (Eq. 11)
        $ans \leftarrow \text{QUERY-FM}(hypo^*)$ 
        $\mathcal{B} \leftarrow \text{FILTER}(\mathcal{B}, hypo^*, ans)$  (Eq. 12)
        $c \leftarrow \text{CONF}(\mathcal{B})$ 
10     $s^K \leftarrow \arg \max_{s_k^* \in \mathcal{B}} w'_k$ 
     $knowledge(\mathcal{B}) \leftarrow s^K$ 
11     $t \leftarrow t + 1$ 
12  until  $t = \text{MAX-STEP}$ ;
13  return  $\pi = (a_0, \dots)$ 
```

update both symbolic predicates and numeric fluents in the KB. This design separates task-level decision making from low-level motion planning. The KB provides a shared representation across these levels. As a result, the PM does not depend on a specific motion planner implementation.

Overall planning pipeline

We adopt an online planning framework as described in Algorithm 1. The planning process starts from the initial knowledge state s_0^K , which is used to initialize the belief (line 3). At each iteration, the planner first selects a_t by performing Monte Carlo tree search over the current belief $\mathcal{B}$ (line 5). If no applicable action exists, the process terminates. Otherwise, the selected action is executed and an observation o_{t+1} is received from the environment.

The belief is then updated using the observation model (line 8), and the confidence of the updated belief is computed (line 8). If the confidence is below the threshold τ , the planner queries the FM to reduce uncertainty. Specifically, it selects the most informative hypothesis, sends it to the FM, and filters out particles inconsistent with the answer (lines 9–9). This process is repeated until the confidence exceeds τ . Once the belief becomes sufficiently confident, the most probable state is committed as the updated knowledge state, and the planner proceeds to the next step.

4.2.3 Sensing and Perception

The SP module acquires raw sensory data, generates local information in $\mathcal{I}^L$, and updates the numeric fluents in the KB. It also provides observations required for belief updates during action execution. Let $\mathcal{M}$ denote the set of perception models. For each predicate pred associated with local information in $\mathcal{I}^L$, there exists a corresponding perception model $m_{\text{pred}} \in \mathcal{M}$ defined as

$$m_{\text{pred}} : \mathcal{R} \rightarrow \mathcal{I}^L,$$

where $\mathcal{R}$ denotes the space of raw sensory data. Given a sensory input $r \in \mathcal{R}$, the model $m_{\text{pred}}(r)$ produces a symbolic fact $f = \text{pred}(\mathbf{x})$ or a numeric fluent $\nu = \langle g(\mathbf{x}), v \rangle$ in $\mathcal{I}^L$.

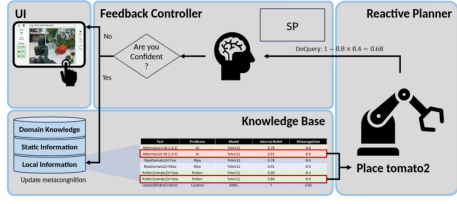
Through this process, perceptual predicates such as `ripe(tomato1)` or `detected(tomato2)` are instantiated, and numeric fluents such as $\nu = \langle \text{distance}(\text{robot}, \text{stem}_1), 2.3 \rangle$ or $\nu = \langle \text{pose.x}(\text{tomato}_1), -0.5 \rangle$ are continuously updated based on incoming observations. Since $\mathcal{I}^L$ represents information that is not known a priori, all elements in this set are generated and refined exclusively through the SP module.

In addition, the SP module defines the observation interface of the system. After executing an action $a \in \mathcal{A}$, the system receives an observation $o \in \mathcal{O}$ derived from the current sensory input, which is used to update the belief over possible states. Furthermore, if triggered by the PM, the SP module acquires new sensory input and evaluates the relevant perception models to produce observations and update the associated elements in $\mathcal{I}^L$.

4.2.4 Feedback Manager

As illustrated in Fig. 4, the FM translates hypotheses provided by the PM into a form that is easily understandable for the user, and interprets user responses into a format that can be processed by the system. Specifically, a hypothesis such as `ripe(tomato1)` is represented internally as a predicate, but the FM converts it into a natural question such as “Is this tomato ripe?”. To support user understanding, the queried object is visualized with a bounding box indicating its location.

In addition, the FM interprets user inputs. Users can respond via touch or speech, and the FM converts these responses into a boolean form (true or false) that can be used for belief updates. Through this bidirectional translation, the FM reduces the semantic gap between the system and the user, enabling efficient interaction.



(a) FM system overview.



(b) User interface overview.

Fig. 4: Overview of the feedback manager and user interface. This figure is a temporary image.

4.2.5 User Interface

As shown in Fig. 4, the UI enables interaction between the user and the system and is implemented on a tablet. It presents queries generated by the FM and overlays the queried object on the camera image using a bounding box to support intuitive understanding. Users provide responses through simple touch input, with optional speech input. The interface hides internal representations and is designed to elicit accurate feedback with minimal workload.

5 Experiments

We evaluate the proposed system through two system-level experiments. First, we analyze how the confidence threshold τ affects task success, execution length, and query frequency. This experiment characterizes the trade-off between autonomous execution and human intervention. Second, we compare four query policies: **All**, **No**, **Random**, and **Ours**. If a query is triggered, we assume that the human provides the correct answer, which isolates the effect of the query policy from user response errors. A separate user study is then used to examine the user-facing effects of these interaction policies.

5.1 Experimental Setup

We evaluate the proposed method in two domains: a simulated waste sorting domain and a real-world tomato harvesting domain. The waste sorting domain is implemented in simulation using NVIDIA Isaac Sim, where a fixed UR5 robotic arm is used for pick and place task. Object detection is performed using the YOLOv12 model [100]. In contrast, the tomato harvesting domain is deployed on a real mobile manipulator consisting of a UR5e robotic arm mounted on a Summit mobile platform. The robot is equipped with an RGB-D camera for visual perception and a LiDAR sensor for localization, and performs object detection using the same YOLOv12.

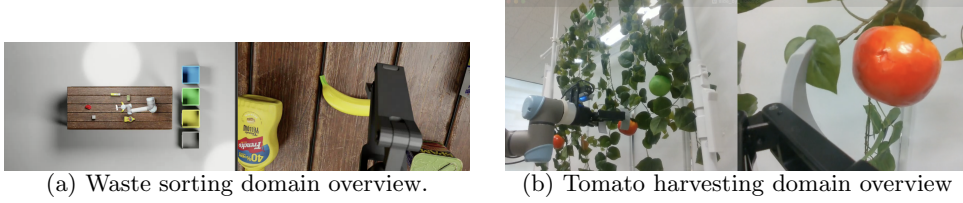


Fig. 5: The illustration of the environments. (a) A UR5 robot sorts four types of waste into designated bins in Isaac Sim. (b) A mobile manipulator harvests ripe tomatoes and discards rotten ones in a real-world setting.

5.1.1 Waste sorting domain

We consider a tabletop waste-sorting scenario with four waste objects placed on a table, as shown in Fig. 5a. A fixed UR5 robot is positioned between the objects and the recycling bins and performs three actions: **detect**, **pick**, and **place**. We assume that all objects are reachable by the manipulator.

Each object belongs to one of four categories: **general**, **paper**, **can**, and **plastic**. The object categories are not known a priori and must be inferred through perception. Since objects may be partially occluded by others, a single observation may not reveal all categories, requiring repeated detection during execution. The **detect** action can observe multiple objects simultaneously, after which the robot places each object into the corresponding recycling bin.

The initial arrangement of objects is randomized in every episode. A task is considered successful when all objects are correctly sorted within the maximum step limit. A reward of +5 is assigned when a waste object is placed into the recycling bin corresponding to its category, and an additional reward of +10 is given when all objects are correctly sorted. All other actions receive zero reward.

5.1.2 Tomato harvesting domain

We consider a tomato harvesting scenario in which four tomatoes are attached to two stems, denoted as **stem_1** and **stem_2**, as shown in Fig. 5b. The robot navigates between three locations: **dock_station**, **stem_1**, and **stem_2**, and performs **detect**, **pick**, **scan**, **place**, and **discard** actions.

Each tomato can be in one of three states: **ripe**, **unripe**, or **rotten**. The tomato states are initially unknown and must be estimated through perception. During detection, ripe and rotten tomatoes are visually ambiguous and cannot always be reliably distinguished. To resolve this ambiguity, the robot performs a **scan** action after picking a tomato to verify its condition before harvesting or discarding it. Unripe tomatoes are left untouched.

The tomato locations are randomized across stems in every episode. A task is considered successful when all ripe tomatoes are harvested and all rotten tomatoes are discarded within the maximum step limit. A reward of +5 is assigned when a ripe tomato is correctly harvested or a rotten tomato is correctly discarded after scanning.

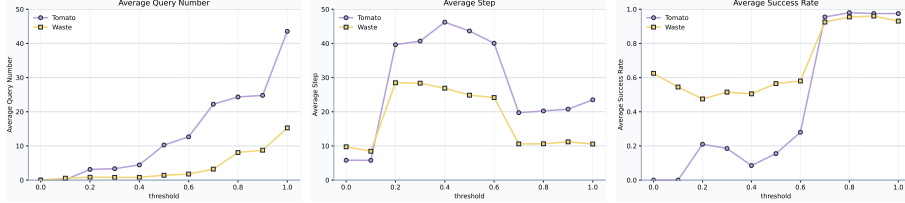


Fig. 6: System performance under different confidence thresholds τ . The threshold controls when the robot requests human feedback. Lower thresholds reduce the number of queries but may leave task-relevant uncertainty unresolved, whereas overly high thresholds increase interaction cost without proportional performance gains.

An additional reward of +10 is given when all task-relevant tomatoes are processed correctly. All other actions receive zero reward.

5.1.3 Implementation Details

In all experiments, the planner used a discount factor $\gamma = 0.95$, UCB exploration constant $c = 1.0$, maximum simulation depth $depth_{\max} = 20$, and $\text{TIMEOUT} = 100$ Monte Carlo simulations per planning step. Each episode was limited to 50 steps. For the policy comparison experiment, the confidence threshold for triggering user feedback was set to $\tau = 0.8$ based on the threshold sweep reported in Sec. 5.2.

5.2 System Evaluation

5.2.1 Experimental Results

We first evaluate the effect of the confidence threshold τ on system performance. Fig. 6 and Table 2 report the average number of queries, execution steps, and task success rate under different threshold values.

As τ increases, the average number of queries generally increases in both domains. In waste sorting, increasing τ from 0.6 to 0.7 changes the success rate from 58.0% to 92.5%, the average number of queries from 1.8 to 3.2, and the average number of execution steps from 24.2 to 10.6. In tomato harvesting, the same change results in a success rate increase from 28.0% to 95.5%, a query count increase from 12.7 to 22.2, and a reduction in execution steps from 40.1 to 19.7. For $\tau \geq 0.7$, the success rate remains above 92% in both domains. At $\tau = 0.8$, the system achieves success rates of 95.5% and 98.0% in waste sorting and tomato harvesting, respectively. The corresponding average numbers of queries are 8.1 and 24.3, and the average numbers of execution steps are 10.6 and 20.2. Based on these results, we use $\tau = 0.8$ in the remaining experiments.

We next compare four query policies at $\tau = 0.8$. **All** requests feedback whenever a query can be generated. **No** performs fully autonomous execution without user feedback. **Random** requests feedback randomly. **Ours** requests feedback when the

Table 2: System evaluation under different confidence thresholds τ . Success rates are reported as percentages.

τ	Waste Sorting			Tomato Harvesting		
	Success Rate (%)	Avg. Queries	Avg. Steps	Success Rate (%)	Avg. Queries	Avg. Steps
0.0	62.5	0.0	9.7	0.0	0.0	5.8
0.1	54.5	0.6	8.4	0.0	0.1	5.8
0.2	47.5	0.8	28.5	21.0	3.1	39.6
0.3	51.5	0.8	28.4	18.5	3.3	40.7
0.4	50.5	0.8	26.9	8.5	4.4	46.2
0.5	56.5	1.4	24.9	15.5	10.3	43.6
0.6	58.0	1.8	24.2	28.0	12.7	40.1
0.7	92.5	3.2	10.6	95.5	22.2	19.7
0.8	95.5	8.1	10.6	98.0	24.3	20.2
0.9	96.0	8.8	11.2	97.5	24.8	20.8
1.0	93.1	15.3	10.6	97.5	43.5	23.5

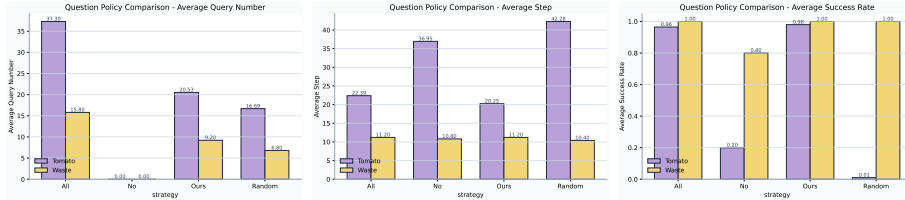


Fig. 7: System performance under different querying strategies. The results compare **All**, **No**, **Random**, and **Ours** in terms of average query count, average execution steps, and task success rate across the two domains.

confidence of the current belief falls below the threshold and selects the query with the highest expected information gain.

Fig. 7 and Table 3 summarize the results. In waste sorting, **All**, **Random**, and **Ours** achieve success rates of 100.0%, whereas **No** achieves 80.0%. The average numbers of queries are 15.8, 6.8, and 9.2 for **All**, **Random**, and **Ours**, respectively. In tomato harvesting, **Ours** achieves a success rate of 98.0%, compared to 96.5% for **All**, 20.0% for **No**, and 1.0% for **Random**. The corresponding average numbers of queries are 20.5, 37.3, 0.0, and 16.7. The average numbers of execution steps are 20.2, 22.4, 36.9, and 42.3, respectively.

5.2.2 Discussion

The results demonstrate that human feedback can improve both task performance and execution efficiency when it is requested selectively. As the confidence threshold increases, the robot acquires additional observations through user feedback, leading to substantial improvements in task success rate. At the same time, the average execution

Table 3: System evaluation under different querying strategies at $\tau = 0.8$. **All** queries whenever feedback is available, **No** performs fully autonomous execution, **Random** queries without using belief confidence, and **Ours** selectively queries based on belief confidence.

Method	Waste Sorting			Tomato Harvesting		
	Success Rate (%)	Avg. Queries	Avg. Steps	Success Rate (%)	Avg. Queries	Avg. Steps
All	100.0	15.8	11.2	96.5	37.3	22.4
No	80.0	0.0	10.8	20.0	0.0	36.9
Random	100.0	6.8	10.4	1.0	16.7	42.3
Ours	100.0	9.2	11.2	98.0	20.5	20.2

length decreases in both domains. For example, increasing τ from 0.6 to 0.7 improves the success rate from 58.0% to 92.5% in waste sorting and from 28.0% to 95.5% in tomato harvesting, while reducing the average execution length by more than half in both domains.

This result suggests that informative observations can guide the planner toward more effective action sequences. If uncertainty remains unresolved, the robot may act on incorrect beliefs, resulting in unnecessary actions or failed execution attempts. By obtaining feedback at uncertain states, the planner can refine its belief earlier and select actions that are more consistent with the underlying state of the environment. Consequently, additional observations can reduce the total number of actions required to complete the task.

The comparison among query policies further highlights the importance of selecting when to request feedback. Although the **All** policy achieves high success rates, it requires substantially more user interactions. In tomato harvesting, **All** requires 37.3 queries on average, whereas **Ours** requires only 20.5 while achieving a slightly higher success rate. A similar trend is observed in waste sorting, where both methods achieve perfect success rates but **Ours** reduces the number of queries from 15.8 to 9.2. These results indicate that many feedback requests generated by continuous supervision provide limited additional information.

In contrast, the **No** policy often fails because belief errors accumulate without correction, while the **Random** policy demonstrates that simply increasing the amount of feedback is insufficient. Feedback must be acquired at states where uncertainty is likely to affect future decisions. By focusing interactions on low-confidence beliefs, the proposed approach maintains high task success, reduces unnecessary user involvement, and shortens execution length.

5.3 User Study

We conducted a Wizard-of-Oz (WoZ) user study to evaluate how different query policies affect user responses, interaction time, subjective workload, and situation awareness during human-robot collaboration. Unlike the system evaluation in Section 5.2, which focused on autonomous task performance, this study focused on the

user-side effect of different interaction strategies. Robot behavior was controlled using predefined scenario scripts, while participants interacted with the system through the same tablet-based interface used by the feedback manager.

5.3.1 Participants

Twelve participants (7 female and 5 male) took part in the study. Participants ranged in age from 21 to 26 years ($M = 23.0$, $SD = 1.48$). Each participant received \$20 upon completion of the study.

5.3.2 Study Design and Conditions

The study employed a within-subject design. Each participant completed ten trials in total, consisting of five waste sorting scenarios and five tomato harvesting scenarios. In each domain, participants experienced five interaction conditions: **All**, **No**, **Ours1**, **Ours2**, and **KnowNo**. **All** requested feedback whenever a query was available, representing continuous human supervision. **No** performed the task without requesting user feedback. **Ours1** used the proposed uncertainty-based state query policy. **Ours2** used a variant of the proposed policy designed to expose participants to a broader range of query situations. **KnowNo** served as an action-level query baseline.

Each domain consisted of five scenarios and five interaction conditions. The assignment between scenarios and conditions was randomized for each participant such that every participant experienced all five conditions exactly once per domain. This design prevented a particular condition from being consistently associated with a specific scenario. Detailed scenario descriptions are provided in Appendix D.

5.3.3 Materials and Procedure

Participants interacted with the robot through a tablet-based interface. The interface displayed the current task state, highlighted task-relevant objects, and presented robot-generated questions together with selectable answer options. The waste sorting scenarios required participants to identify the category of waste objects, including general waste, paper, cans, and plastic. The tomato harvesting scenarios required participants to identify the state of tomatoes, such as whether a tomato was ripe, unripe, or rotten.

Before the experiment, participants received an explanation of the study objectives, the two task domains, and the robot interaction process. They were then introduced to the tablet interface and completed practice interactions to become familiar with the available information and response procedure. During the main session, participants completed ten trials without assistance. If a participant selected an incorrect answer, the response was recorded as incorrect for evaluation purposes, but the scenario continued using the intended task state so that subsequent interactions were not affected by earlier mistakes. After each scenario, participants completed a questionnaire regarding the interaction experience. After completing all ten trials, participants

completed post-study questionnaires and provided qualitative feedback during a short debriefing session.

5.3.4 Measures

We organized the user-study analysis along three axes. First, we evaluated response quality using scenario-condition answer accuracy and task success. Answer accuracy measures whether each user response matched the predefined answer key. Task success is stricter: a task is counted as successful only when all required answers in that task were correct. Because **No** does not request user answers, it is not scored in this response-quality analysis.

Second, we measured time cost. Operation time was computed from the recorded timeline after subtracting the scenario video duration, so that the value reflects pure user operation time rather than passive viewing time. Survey time was measured separately. Missing scenario-condition cells were retained as empty cells in the summary tables and excluded from mean calculations.

Third, we analyzed subjective and situation-awareness measures. Subjective workload was measured using NASA Raw-TLX. Situation awareness was evaluated using SAGAT-style questions and SART. Cognitive fatigue was measured using a self-reported fatigue item. Unless otherwise specified, questionnaire responses were recorded using a 5-point Likert scale. For the within-subject questionnaire analysis, we used Friedman tests across the five conditions and planned exact Wilcoxon signed-rank tests centered on **Ours1** and **Ours2**, with Holm correction within each measure and domain panel.

5.3.5 Results: Response Accuracy and Task Success

Fig. 8 summarizes the response-quality results across both domains. In answer-level accuracy, **All**, **Ours1**, and **Ours2** generally remained high across scenarios, whereas **KnowNo** decreased in the later scenarios. For example, in Scenario 4, answer accuracy was 98.5% for **All**, 89.7% for **Ours1**, 97.5% for **Ours2**, and 66.7% for **KnowNo**. In Scenario 5, **Ours1** reached 100.0%, **Ours2** reached 93.3%, and **KnowNo** reached 73.3%.

The stricter task-success metric shows the cost of individual mistakes more clearly. **Ours2** achieved perfect task success in Scenarios 1–3 and remained high in Scenario 4 (85.7%), while **KnowNo** dropped to 16.7% in Scenario 4 and 0.0% in Scenario 5. **Ours1** showed perfect task success in Scenarios 1 and 5, but decreased in Scenario 4 (50.0%). These results suggest that state-level queries can support accurate user responses, but the benefit depends on the scenario and on how the query distribution interacts with task difficulty.

5.3.6 Results: Operation and Survey Time

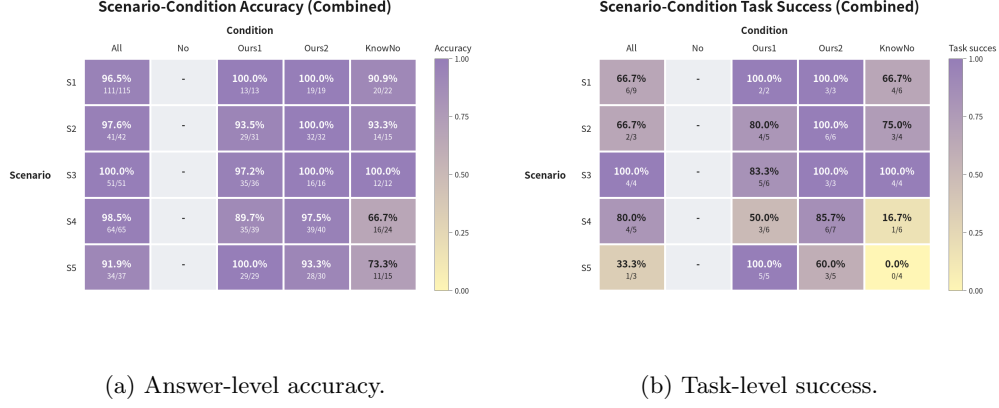


Fig. 8: Response-quality results by scenario and condition. Answer accuracy scores individual responses, whereas task success counts a task as successful only if all required answers were correct. Blank cells indicate conditions that did not produce scorable user responses.

Fig. 9 summarizes pure operation time and survey time for the two domains. After subtracting scenario video duration, **No** had the shortest pure operation time, as expected, because it did not require user responses. In waste sorting, the mean pure operation time across available scenarios was 0.44 s for **No**, 6.90 s for **Ours1**, 7.87 s for **Ours2**, 20.31 s for **KnowNo**, and 22.19 s for **All**. In tomato harvesting, the corresponding values were 1.78 s for **No**, 14.27 s for **Ours1**, 13.98 s for **Ours2**, 25.33 s for **KnowNo**, and 23.60 s for **All**.

The time results separate two effects. First, selective state-level querying reduced operation time relative to continuous querying and the action-level baseline. In both domains, **Ours1** and **Ours2** required less pure operation time than **All** and **KnowNo**. Second, survey time did not follow the same ordering as operation time. In waste sorting, the average survey time was 68.53 s for **Ours1**, 76.76 s for **KnowNo**, 91.10 s for **All**, 94.08 s for **Ours2**, and 107.75 s for **No**. In tomato harvesting, **Ours2** (65.16 s) and **Ours1** (66.47 s) were shorter than **All** (76.45 s), **KnowNo** (89.00 s), and **No** (91.63 s). This suggests that reducing interaction during execution does not necessarily reduce post-trial reporting time; participants may still need time to reconstruct what happened when the robot did not ask questions.

5.3.7 Results: Situation Awareness and Subjective Burden

Fig. 10 summarizes the questionnaire results that were most relevant to the proposed query policy. In the waste sorting domain, SAGAT accuracy differed significantly across conditions, $\chi^2(4) = 12.92$, $p = .012$, Kendall's $W = 0.27$. **Ours1** had the highest mean accuracy ($M = 0.72$, $SD = 0.15$), compared with **All** ($M = 0.56$, $SD = 0.19$), **No** ($M = 0.67$, $SD = 0.07$), **Ours2** ($M = 0.53$, $SD = 0.12$), and **KnowNo** ($M = 0.60$, $SD = 0.24$). The planned comparison showed that **Ours1** was

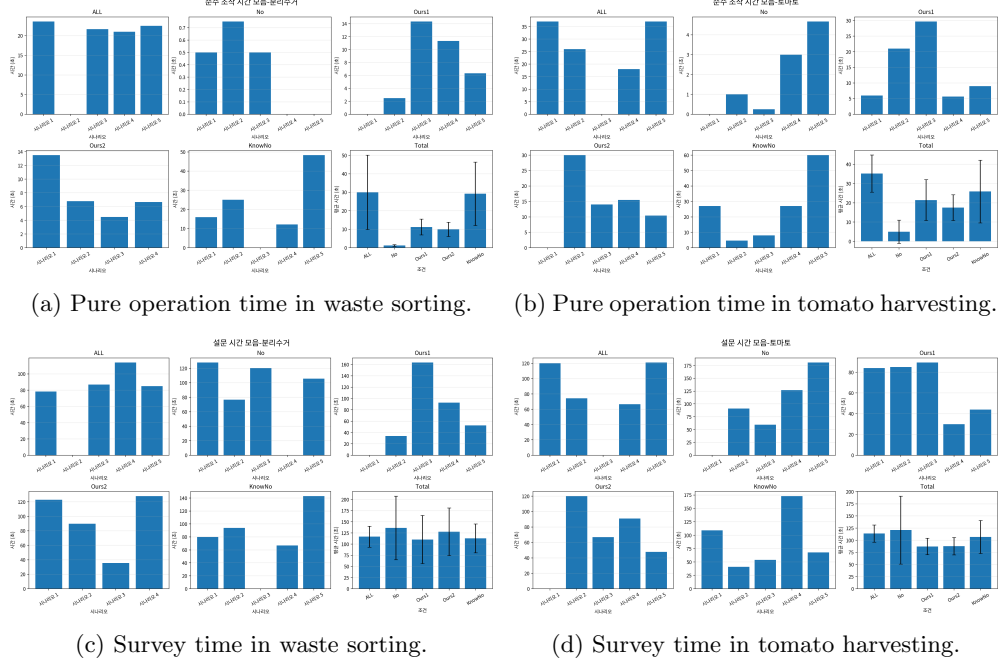
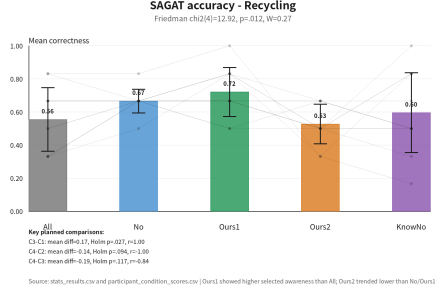


Fig. 9: Time-cost results by domain. Operation time subtracts the scenario video duration and therefore reflects pure user operation time. Survey time is reported separately. Empty scenario-condition cells in the source data are retained as missing values and excluded from means.

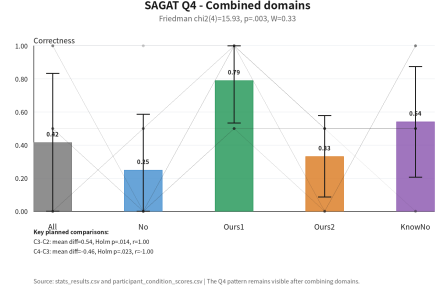
significantly higher than **All** (mean difference = 0.17, Holm-adjusted $p = .027$, rank-biserial $r = 1.00$). In tomato harvesting, the overall SAGAT condition effect was not significant, $\chi^2(4) = 4.06$, $p = .398$, $W = 0.08$.

The strongest situation-awareness signal appeared in SAGAT Q4, which asked participants to identify why the robot requested information. In waste sorting, Q4 accuracy differed strongly across conditions, $\chi^2(4) = 22.12$, $p < .001$, $W = 0.46$. **Ours1** reached perfect Q4 accuracy ($M = 1.00$, $SD = 0.00$), outperforming **No** ($M = 0.17$, $SD = 0.39$; Holm-adjusted $p = .012$, $r = 1.00$). **Ours2** was lower than **Ours1** ($M = 0.08$ vs. 1.00 ; Holm-adjusted $p = .007$, $r = -1.00$). The same Q4 pattern remained visible after combining both domains, $\chi^2(4) = 15.93$, $p = .003$, $W = 0.33$: **Ours1** exceeded **No** (mean difference = 0.54, Holm-adjusted $p = .014$), while **Ours2** was lower than **Ours1** (mean difference = -0.46 , Holm-adjusted $p = .023$).

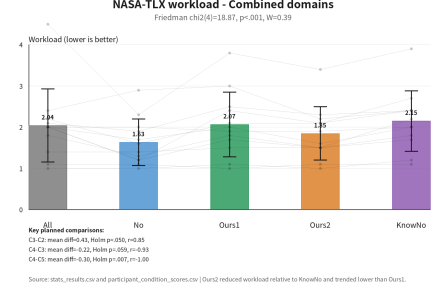
Subjective burden showed a different pattern. NASA-TLX workload differed significantly across conditions in the aggregated data, $\chi^2(4) = 18.87$, $p < .001$, $W = 0.39$. **Ours2** produced lower workload ($M = 1.85$, $SD = 0.64$) than **KnowNo** ($M = 2.15$, $SD = 0.73$; Holm-adjusted $p = .007$, $r = -1.00$), and also trended lower than **Ours1** ($M = 2.07$, $SD = 0.79$; Holm-adjusted $p = .059$, $r = -0.93$). Effort followed the same direction, $\chi^2(4) = 21.86$, $p < .001$, $W = 0.46$. **Ours2** required less effort than **All**



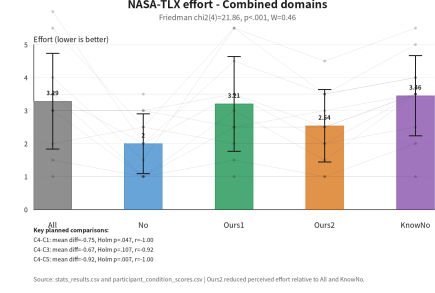
(a) SAGAT accuracy in waste sorting.



(b) SAGAT Q4 across both domains.



(c) NASA-TLX workload across both domains.



(d) NASA-TLX effort across both domains.

Fig. 10: Situation-awareness and subjective-burden results. The panels report the questionnaire measures that showed the clearest condition-dependent patterns in the user study.

($M = 2.54$ vs. 3.29 ; Holm-adjusted $p = .047$, $r = -1.00$) and **KnowNo** ($M = 2.54$ vs. 3.46 ; Holm-adjusted $p = .007$, $r = -1.00$). In waste sorting, **Ours2** also reduced effort relative to **KnowNo** ($M = 2.58$ vs. 4.00 ; Holm-adjusted $p = .014$, $r = -1.00$).

5.3.8 Discussion

The three analysis axes show that query timing and query content affect different aspects of the interaction. The response-quality analysis indicates that state-level queries can support accurate answers, especially when evaluated at the stricter task level. **Ours2** achieved perfect task success in the first three scenarios and remained substantially higher than **KnowNo** in the later scenarios. This result is consistent with the planning motivation of the proposed method: instead of asking the user to choose the robot's next action, the robot asks about a state hypothesis that resolves task-relevant uncertainty.

The time analysis shows the practical cost of each interaction strategy. As expected, **No** minimized operation time because the user had almost nothing to do during execution. This does not make **No** a desirable collaborative strategy, because it removes the user’s ability to correct robot uncertainty. Among the interactive conditions, **Ours1** and **Ours2** reduced pure operation time relative to **All** and **KnowNo** in both domains. This suggests that selective state-level querying can reduce the direct time cost of human intervention compared with continuous supervision and action-level questioning.

The questionnaire analysis clarifies the awareness–burden tradeoff. **Ours1** produced the clearest situation-awareness benefit, especially for SAGAT Q4. This suggests that uncertainty-triggered state queries can make the reason for robot uncertainty more legible to the user. However, **Ours1** did not reduce workload as consistently as **Ours2**. In contrast, **Ours2** reduced workload and effort relative to **KnowNo**, but did not reproduce the same Q4 advantage as **Ours1**. Thus, the two variants should not be interpreted as a simple winner-loser comparison. **Ours1** emphasizes selected awareness at critical uncertainty points, whereas **Ours2** distributes queries across a broader set of situations and produces lower subjective burden.

These results support the central claim that human feedback should be treated as a limited resource in robot planning. Asking at every opportunity can preserve supervision but increases interaction cost. Asking no questions minimizes immediate burden but gives the robot no opportunity to repair uncertain beliefs. Asking action-level questions can shift planning responsibility to the user. The proposed state-level querying approach offers a middle ground: it asks only when belief uncertainty is high and frames the question around the state information needed by the planner.

The study also has limitations. The sample size was small, and several effects should be interpreted as descriptive or exploratory even when the direction is consistent. The SAGAT benefit of **Ours1** was concentrated in waste sorting and especially in Q4, so it should be described as an improvement in selected awareness measures rather than a general improvement in all aspects of situation awareness. Finally, the WoZ protocol controlled the interaction sequence and isolated the user-side effect of the query policy; deployment in a fully autonomous system requires the system-level reliability evaluated separately in Section 5.2.

6 Conclusion and Future Work

In this paper, we proposed a robotic decision-making framework that integrates symbolic knowledge and uncertainty, with the goal of reducing user workload. The proposed method maintains a belief over possible worlds within a POMDP-based planning framework and quantitatively models uncertainty through a probability distribution over these worlds. Based on this, the robot can assess its own uncertainty and selectively query the human only when necessary. This approach enables stable task execution without continuous supervision and minimizes unnecessary interactions, thereby reducing user intervention.

In particular, we leverage entropy-based confidence to determine when to issue queries and select the most informative hypothesis to construct questions, improving the efficiency of interaction. Experimental results show that the proposed method reduces the number of human interactions while maintaining or improving task success rates, and effectively lowers user workload. User studies further support this trend, demonstrating that a system which actively manages uncertainty and requests assistance only when needed can operate effectively in real-world settings.

For future work, several extensions can be considered. First, since the current query mechanism is based on symbolic hypotheses, it can be extended to a more intuitive natural language interface to improve user experience. In addition, incorporating more refined models of perception uncertainty would improve the accuracy of uncertainty estimation. Furthermore, this approach can be extended to multi-robot settings, where multiple agents share uncertainty and coordinate their queries collaboratively.

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Appendix Contents

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Appendix A Planning Algorithms

This appendix provides the detailed pseudocode for the POMCP-based search routines described in Sec. 4.2.2.

Appendix B Detailed Experimental Settings

Appendix C Experimental Results

Appendix D User Study

Algorithm 2: POMCP Search with Applicable Actions

```
1 Input: belief  $b$   
   Output: selected action  $a \in \mathcal{A}$   
2 procedure SEARCH( $b$ )  
3    $h \leftarrow$  root history  
    $s^K \leftarrow \text{knowledge}(b)$   
  
4   repeat  
5     if  $B(h) \neq \emptyset$  then  
6       sample  $s \sim B(h)$   
7     else  
8        $s \leftarrow s^K$   
9     SIMULATE( $s, h, 0$ )  
10  until TIMEOUT;  
11   $a \leftarrow \text{SEARCH-BEST}(h, s^K, \text{false})$   
   return  $a$ 
```

Algorithm 3: POMCP Simulation with Applicable Actions

```

1 Input: state  $s$ , history  $h$ , depth  $depth$ 
   Output: simulated return  $R$ 
2 procedure SIMULATE( $s, h, depth$ )
3   if  $\gamma^{depth} < \epsilon$  then
4     return 0
5   if there is no  $a[s] \in \mathcal{A}$  then
6      $N(h) \leftarrow N(h) + 1$ 
7     return 0
8   if  $h \notin T$  then
9     foreach  $a[s] \in \mathcal{A}$  do
10       $T, N(ha), V(ha) \leftarrow T \cup \{ha\}, 0, 0$ 
11       $N(h) \leftarrow N(h) + 1$ 
12      return ROLLOUT( $s, depth$ )
13   $a \leftarrow \text{SEARCH-BEST}(h, s, \text{true})$ 
14  sample  $(s', o, r)$  from simulator  $\mathcal{G}(s, a)$ 
15   $R \leftarrow r + \gamma \cdot \text{SIMULATE}(s', hao, depth + 1)$ 
16   $B(h) \leftarrow B(h) \cup \{s\}$ 
17   $N(h) \leftarrow N(h) + 1$ 
18   $N(ha) \leftarrow N(ha) + 1$ 
19   $V(ha) \leftarrow V(ha) + \frac{R - V(ha)}{N(ha)}$ 
20 return  $R$ 

```

Algorithm 4: Action Selection and Rollout

```
1 Input: history  $h$ , state  $s$ 
2 procedure SEARCH-BEST( $h, s, use\_ucb$ )
3    $\mathcal{C} \leftarrow \{a \mid a[s] \in \mathcal{A}, ha \in T\}$ 
4   if  $\mathcal{C} = \emptyset$  then
5     return  $\emptyset$ 
6   if  $use\_ucb$  then
7     return  $\arg \max_{a \in \mathcal{C}} V(ha) + c \cdot \sqrt{\frac{\log N(h)}{1+N(ha)}}$ 
8   else
9     return  $\arg \max_{a \in \mathcal{C}} V(ha)$ 
10 procedure ROLLOUT( $s, depth$ )
11   if  $(\gamma^{depth} < \epsilon \vee \gamma = 0) \wedge depth > 0$  then
12     return 0
13   if there is no  $a[s] \in \mathcal{A}$  then
14     return 0
15    $a \sim \text{Uniform}(\{a \mid a[s] \in \mathcal{A}\})$ 
16   sample  $(s', o, r)$  from simulator  $\mathcal{G}(s, a)$ 
17   return  $r + \gamma \cdot \text{ROLLOUT}(s', depth + 1)$ 
```
